



ATLAS CAMERA · SHOWCASE DEEP-DIVE 02 / 11 [← ALL WORKFLOWS](#)

CITY BLOCKS — AERIAL PRESET

A real DSLR aerial through the one-node aerial preset, conformed and handed to Maya.

FILE `atlas_city_blocks_newyork_workflow.json` VERIFIED 2026-07-16

THE STORY

This is the set's real-photography proof: a 177 MB, 7360×4912 aerial frame — not an AI plate. GeoCalib's gravity is trustworthy on real glass, so no roll trim is needed. The `scene_type=aerial` preset resolves one node into `geometry_mode=both + azimuth_walls + max_objects=6`: buildings become fitted boxes over a relief ground and treeline.

The one thing real photography does NOT fix is metric scale from a high vantage: the street-level ground fit fails (cars break the plane) and the solve falls back to the 1.6 m default. The fix is the sky-rise doctrine — **count the levels**. The 'PALM TOO' tenement mid-frame is 5 storeys × 3.5 m = 17.5 m; its bounding box plus that counted height feed `AtlasReferenceScaleSolve`, and single-view geometry locks tier-1 metric scale: camera at 63.7 m (0.96 confidence, photographer-confirmed high vantage). Every downstream number is now measured, not dialed.

IN THE VIEWPORT — PROJECTED VS. THE GREY MESH



SOURCE PLATE

newyork_Birdseye.png · real photo · 7360×4912 · 177 MB — the single still photograph everything below is reconstructed from.



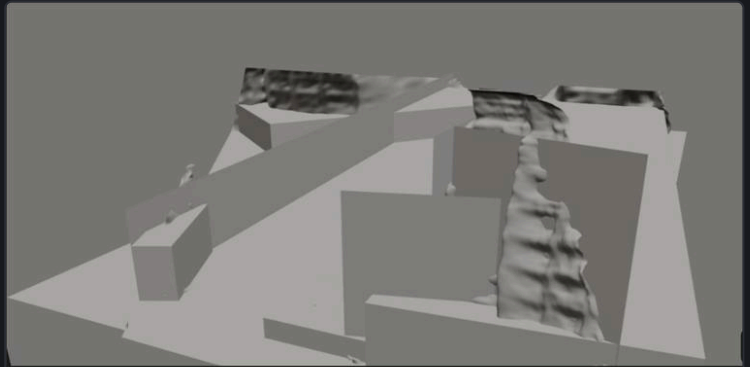
PROJECT ON

The photo reassembled on the recovered camera — texels assigned by ray, perfect from this view.



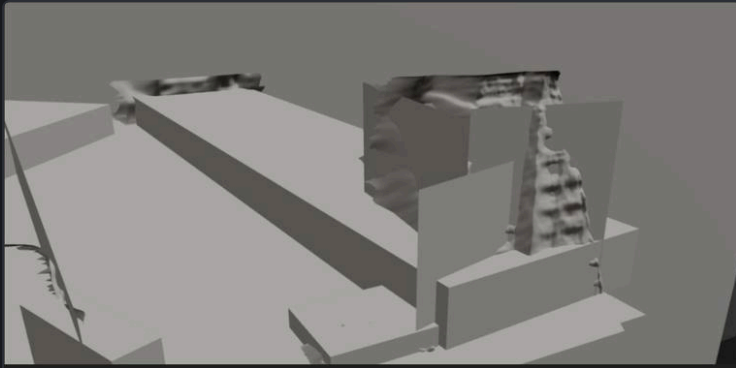
PROJECT OFF

The same geometry as a grey mesh, recovered view — what the depth-derived surfaces actually are.



GREY · ORBIT LEFT

Orbited off-axis: the reconstructed cone. Coverage ends where the camera never saw.



GREY · ORBIT RIGHT

The opposite angle — silhouette tears read as deliberate holes, not errors.

NODES ON STAGE

AtlasLearnedSolveFromImage

AtlasReferenceScaleSolve

AtlasDeriveProjectionGeometry

AtlasDefineShotCam

AtlasExportMayaReviewScene

AtlasExportReviewPackage

AtlasExportReliefMesh

RETUNE ON YOUR PLATE

- **Count the storeys — the sky-rise scale doctrine.** Pick a building fully visible base-to-roof, count its levels ($\times 3.5$ m), draw its bbox into AtlasReferenceScaleSolve with that height. Tier-1 scale by geometry beats any eyeballed dial — the original 25 m guess here was $\sim 2.5\times$ small against the photographer's ground truth.
- **AtlasDefineShotCam is your delivery format.** Sensor mm + focal + long-edge resolution — a Nuke/Resolve-style project format that conforms the viewport render without touching how the photo projects.
- **max_objects on the derive node.** Six boxes read clean on this plate; dense downtown frames may want more, sparse suburbs fewer.

RUN-VERIFIED

RUN ✓ 7 fitted primitives + relief at counted real-world metres (camera 63.7 m, scale_source=reference_object); viewport conformed to 1920 $\times$ 1280; Maya review scene (with the real relief OBJ) and the full review package both written.

REQUIREMENTS

Atlas + [neural] · heavy plate (browser preview slow; server fine) · no gates

